

Task Overview

Build a background job processing API. Clients should be able to submit jobs, track their progress, and retrieve results. Jobs are processed asynchronously by workers running in the background.

You decide the architecture. We care about your reasoning as much as your implementation.

Requirements

Core API

- Submit a job and get an ID back immediately
- Poll a job by ID, return its current status and result when done
- List jobs with basic filtering

Workers

- At least two distinct job types with real processing logic
- Retry failed jobs with backoff
- Control how many jobs run concurrently

Infrastructure

- Queue-backed
- Persistent state
- The whole system starts with a single command

Auth

- Protect the API appropriately

Deliverables

Item	Expectation
Codebase	Clean, runnable Git repository. docker-compose up should bring everything up cold with no manual steps.
Tests	Good coverage. Tests should pass out of the box.
Document	A short document covering your architecture decisions and a system design diagram showing how the components fit together.

AI Usage

You may use AI tools freely. We ask one thing: include a short section in your document called AI Usage.

In it, describe:

- What you used AI for
- Where you pushed back on or discarded AI output
- One decision where you chose a different approach than AI suggested, and why

Notes

We will not be reading this for line count. We will be reading it for judgment. Make decisions, own them, and be ready to defend them.

If something is intentionally left out or simplified, say so in your document.

Good Luck.